

VeriID AI — Final Project State & Handover Document

Version: Day 4 Final Integration Completed | Benchmark: 100.00% Pass Rate (20/20 Samples)

1. Project Overview & Multi-Modal Architecture

VeriID AI evaluates identity documents against live facial captures across forensic risk vectors:

- **OCR Engine:** Dynamic EasyOCR tokenization, fuzzy keyword classification, regex ID parsing, and chronological validity checks.
- **Forensic ELA Engine:** Dual-scale JPEG Error Level Analysis (Q90/Q75) with contour thresholding and localized bounding boxes.
- **Biometrics & PAD:** DeepFace (Facenet512) 1:1 facial verification with passive FFT Moiré spectrum, glare ratio tracking, and execution latency profiling.
- **Multi-Factor Risk Engine:** Weighted risk score aggregation with three-tier verdict thresholds.

2. Final Repository Structure

veriid-ai/

- app.py — Streamlit Risk Dashboard & Forensic Heatmap Visualizer
- src/
 - ■■■■ ocr_engine.py — EasyOCR + Regex validation + Fuzzy classification + Token confidence
 - ■■■■ ela_engine.py — Dual-scale ELA (Q90/Q75) + Bounding box localization
 - ■■■■ face_engine.py — DeepFace Facenet512 + FFT PAD + Latency telemetry
 - ■■■■ risk_engine.py — Weighted scoring & verdict assignment
- scripts/
 - ■■■■ run_benchmark.py — 20-sample automated evaluation matrix
 - ■■■■ generate_handover_pdf.py — Project state report generator
- data/test_samples/ — 20 ground-truth verified test image pairs & stress tests

3. Engine Configurations & Calibrated Thresholds

Engine / Vector	Model / Method	Threshold / Metric	Risk Weight
Biometric Match	Facenet512 (Cosine)	Distance <= 0.25 (Sim >= 85%)	60.0%
Anti-Spoofing	FFT Moiré + Glare	Spoof Score < 50.0	60.0%
Forensic ELA	Dual-Scale ELA (Q90/Q75)	Anomaly > 5.0 or Boxes > 0	40.0%
OCR Syntax	EasyOCR + Date Regex	Format Valid & Age >= 18	30.0%

4. Final Benchmark Performance (scripts/run_benchmark.py)

- **Total Evaluated:** 20 Samples (Genuine, Tampered, Impersonation, Screen Spoof, Underage, Skewed, High Glare, Blur, Format Anomaly, Print Attack)
- **Accuracy Rate:** 100.00% (20/20 Passed)
- **False Acceptance Rate (FAR):** 0.00% (Target: < 5%)
- **False Rejection Rate (FRR):** 0.00% (Target: < 5%)

5. Day 4 Completed Deliverables (All PRs Merged to main)

- **Member 1 (PR #6):** Fuzzy token classification and per-field confidence scoring — **Merged**
- **Member 2:** Multi-vector risk aggregation and three-tier decision engine — **Complete**
- **Member 3 (PR #5):** Multi-channel PAD & execution latency profiling — **Merged**
- **Member 4 / Lead (PR #7):** Dual-scale ELA & localized bounding boxes — **Merged**
- **Member 5:** 20-sample dataset expansion & edge-case stress suite — **Merged**